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APPARATUS FOR PROVIDING AN ELECTRONIC UNIT WITH A WATER SEAL, EMI SHIELD, AND EMI CONTACTS

Abstract

An apparatus comprising an electronic unit is provided. The electronic unit may include a housing having a cover and a base that define an opening for receiving a printed circuit board, and a plurality of electronics positioned thereon that may generate electromagnetic interference (EMI) and be susceptible to water damage. A first gasket dispensed onto the cover is configured to at least partially remove electromagnetic interference generated from the plurality of electronics and to prevent liquid from entering into the opening of the housing. A second gasket dispensed onto the base is configured to prevent liquid from penetrating into the opening of the housing. A connector support fitted between the cover and the base may include a first rib that presses into the second gasket to prevent liquid from penetrating into the housing, and a second rib that bites into the base of the housing to provide EMI contact.

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Background/Summary

TECHNICAL FIELD

[0001] Aspects disclosed herein generally relate to electronic units with a housing for use in establishing a fluid seal, an electromagnetic interference (EMI) shield, and EMI contacts for electronic units. These aspects and others will be discussed in more detail below.

BACKGROUND

[0002] Vehicle electronics have steadily increased in complexity since the introduction of engine control units in the early 1970s. With that, comes a desire for finding innovative ways to house electronic units installed in vehicles, for example, so as to protect and enhance the overall function of a unit.

SUMMARY

[0003] In at least one aspect, an electronic unit is provided. The electronic unit may comprise a printed circuit board (PCB); a plurality of electronics positioned on the PCB; a housing including a cover and a base that define an opening for receiving the PCB and the plurality of electronics; and a first gasket encircling an outer periphery of the PCB and being configured to at least partially remove electromagnetic interference generated from the plurality of electronics and to prevent liquid from exterior of the housing from penetrating into the opening of the housing.

[0004] In another aspect, an electronic unit is provided. The electronic unit may comprise a printed circuit board (PCB); a plurality of electronics positioned on the PCB; a housing including a cover and a base that define an opening for receiving the PCB and the plurality of electronics; and a first gasket positioned on an outer periphery of the PCB and being configured to at least partially remove electromagnetic interference generated from the plurality of electronics and to prevent liquid from exterior of the housing from penetrating into the opening of the housing.

[0005] In yet another aspect, an electronic unit is provided. The electronic unit may comprise a printed circuit board (PCB); a plurality of electronics positioned on the PCB; a housing including a cover and a base that define an opening for receiving the PCB and the plurality of electronics; and a connector support to receive a plurality of connectors. The connector support may be positioned between the cover and the base and may further be positioned on a side of the housing. The cover of the housing may include a first gasket positioned on an outer periphery of the PCB. The gasket may be configured to at least partially remove electromagnetic interference generated from the plurality of electronics and to prevent liquid from exterior of the housing from penetrating into the opening of the housing. The base of the housing may include a second gasket positioned on an outer perimeter of the base on the side of the housing. The second gasket may be configured to prevent liquid from exterior of the housing from penetrating into the opening of the housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 illustrates an electronic unit including a housing in accordance with an embodiment.

[0007] FIG. 2 illustrates an exploded view of the electronic unit of FIG. 1 in accordance with an embodiment.

[0008] FIG. 3 illustrates a cover and a corresponding electromagnetic compatibility (EMC) gasket in accordance with an embodiment.

[0009] FIG. 4 illustrates the EMC gasket dispensed onto the housing cover in accordance with an embodiment.

[0010] FIG. 5 illustrates the PCB secured to the base of the housing in accordance with an embodiment.

[0011] FIG. 6 illustrates a top view of the base of the housing with a second gasket being dispensed onto the base in accordance with an embodiment.

[0012] FIG. 7 illustrates a top view of the electronic unit including a connector support with a first rib for promoting the sealing of the connectors in accordance with an embodiment.

[0013] FIG. 8 illustrates a detailed view of the connector support of the housing and the connectors in accordance with an embodiment.

[0014] FIG. 9 illustrates the manner which the housing is assembled in accordance with an embodiment.

[0015] FIG. 10 illustrates another view in which the base, the cover, and a connector assembly are assembled with one another to form the housing in accordance with an embodiment.

[0016] FIG. 11 illustrates another view in which the base, the cover, and the connector assembly are assembled with one another to form the housing in accordance with an embodiment.

DETAILED DESCRIPTION

[0017] As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention that may be embodied in various and alternative forms. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.

[0018] Provided herein is an electronic unit including a housing for providing a water seal, an EMI shield, and EMI contact in a single part. The electronic unit may be a controller. Commonly, a water seal for connectors is achieved by providing an individual water seal around each connector. This approach, however, does not provide enough of a water seal. Often seals of this nature fail to pass an IPX8 test due to a tolerance stack up around each connector. Additionally, this approach does not provide enough of the EMI shield and the EMI contact (i.e., grounding) for connectors. Using this approach, a separate EMI shield may be needed for the connectors.

[0019] The electronic units provided herein in accordance with at least some embodiments may include a housing that combines a water seal that passes IPX8 standards (e.g., water immersion tests) for water resistance, an EMI shield, and EMI contact in one single part. Housings disclosed in some embodiments may eliminate the tolerance issue that occurs when individual water seals are provided around each connector, provide the robust EMI shield and the EMI contact (e.g., grounding) for connectors, and achieve an IPX8 rating.

[0020] In general, the disclosed housing may be utilized in connection with any electronic unit that requires a water seal (including an IPX8 waterproof rating), a robust EMI shield and an EMI contact for connectors. In an example, the disclosed housing may be utilized in connection with electronics units in an automotive vehicle. Non-limiting examples of the vehicle electronic units (or central control units) may include but not limited to, infotainment head units, instrument panel modules (or instrument panel control units), navigation modules (or navigation control units), audio control units, on-board charging devices, etc.

[0021] IPX8 testing is used to determine the ability of an electronic unit or enclosure to withstand fluid (e.g., water) submersion for extended periods of time. The test standard describing the details of IPX8 testing is IEC 60529. In one example, to achieve an IPX8 waterproof rating, an electronic unit or enclosure is required to remain submerged in, for example, 1 meter of water for up to 30 minutes without leakage or malfunction. The housing disclosed herein may meet IPX8 waterproofing standards.

[0022] While EMI shielding may be achieved by utilizing any number of electrical/electronic mechanisms to alter the frequency, such mechanisms may be costly and less robust. The mechanical shielding offered by the housing provided herein may reduce the emittance of EMI and electronic noise in a more robust and cost-effective manner.

[0023] FIG. 1 illustrates a top view of an electronic unit **100** in accordance with one embodiment as will be further described herein. The electronic unit **100** may include at least one controller (not shown) that is positioned on the PCB **112** and may be part of the electronics **113**. The controller may be configured to control any number of operations in a vehicle. The electronic unit **100** generally includes a housing **102**. In one example, the housing **102** may undergo die casting and be formed of metal. Further, the housing **102** may be formed of various non-ferrous metals such as, but not limited to, zinc, copper, aluminum, magnesium, lead, pewter, tin-based alloys, etc. The housing **102** may alternatively be formed of ferrous metals or plastics. The housing **102** may include a base **104**, and a cover **106** sized to be fitted over the base **104**. The housing **102** may further include a connector support **108** sized to fit at least one connector **110a**, **110b**, **110c**, **110d**, **110e** ("**110**"). The electronic unit **100** may further include a printed circuit board **112** (see FIG. 2) including a plurality of electronics **113** positioned thereon. The PCB **112** along with the plurality of electronics **113** are positioned within an opening **105** of the housing **102**. The plurality of electronics **113** of the electronic unit **100** are susceptible to generating electromagnetic interference (EMI) and noise.

[0024] FIG. 2 illustrates an exploded view of an electronic unit **100** including the housing **102** in accordance with an embodiment. As shown, the housing **102** may be formed as multiple parts including the base **104**, the cover **106**, and the connector support **108**. In one example, the PCB **112** including the electronics **113** may be secured to the connector support **108**. The PCB **112** may be sized to be fitted into a first recess **114** in the base **104** of the housing **102**. The connector support **108** may be sized to be fitted into a second recess **116** in the base **104** of the housing **102**. A wall (or divider) **132** separates the first recess **114** from the second recess **116**. The electronic unit **100** may be assembled by fitting the connector support **108** into the second recess **116** of the base **104** and the PCB **112** into the first recess **114** of the base **104** and securing the connector support **108** and the PCB **112** to the base **104** with fasteners **118**. In one example, the fasteners **118** may be screws. The cover **106** may be positioned over the base **104**. The cover **106** may be attached to the base **104** using the fasteners **118**. It is recognized that the cover **106** and the base **104** may have apertures **120** through which the fasteners may be inserted. It is also recognized that the cover **106** may be secured to the base **104** via any number of male-female interfaces or configurations. It is also further recognized that the PCB **112** may also have apertures **121** through which the fasteners may be inserted. Alternatively, the fasteners **118** may be drilled or pressed through the PCB **112** to secure the PCB **112** to the base **104** of the housing **102**.

[0025] FIG. 3 illustrates an exploded view of the cover **106** of the housing **102** and a first gasket **122**. The first gasket **122** may be an electromagnetic compatibility gasket (EMC gasket). The first gasket **122** may aid in mitigating EMI that may be generated by the plurality of electronics **113** and also aid in preventing water from intruding into housing **102**. The first gasket **122** may also reduce pressure or tension on the PCB **112** when the fasteners **118** are attached to the cover **106** to secure the cover **106** to the base **104**. The first gasket **122** may be fitted along an interior of the perimeter of the cover **106**. In an example, the first gasket **122** may be dispensed onto a bottom surface **124** of the cover **106** and cured in place. In an example, the first gasket **122** may be dispensed onto the cover by a computer numerical control (CNC) machine. It is recognized that other methods for dispensing the first gasket **122** onto the cover may be used as well. In an example, the first gasket **122** may be formed from a rubber or a silicone dispensable material. In an example, the first gasket **122** may be formed from a rubber or silicone dispensable material that also includes metal to promote metal to metal contact between the base **104** of the housing **102** and the cover **106** of the housing **102**. Alternatively, the rubber or silicone gasket may be mixed with a metal powder. The

metal powder in the first gasket **122** may provide electric current grounding for the base **104** and the cover **106**. In addition to being dispensed along the perimeter of the bottom surface **124** of the cover **106**, the first gasket **122** may be dispensed along the perimeter of each of the apertures **120** through which the fasteners **118** may be inserted. In this way, the first gasket **122** may provide a water seal around the area surrounding the fasteners **118** to prevent water (liquid) from intruding therethrough. The cover **106** may further include guide pins **126a**, **126b**, **126c**, and **126d** (“**126**”). The guide pins **126** may project outwardly from the bottom surface **124** of the cover **106** to aid in aligning the cover **106** with the base **104** during the assembly of the housing **102**.

[0026] FIG. **4** illustrates the first gasket **122** being dispensed onto the cover **106** of the housing **102** in accordance with an embodiment. The first gasket **122** is dispensed onto the bottom surface **124** of the cover **106** and is dispensed along the perimeter of the cover **106** and along the perimeter of the apertures **120** through which the fasteners **118** may be inserted.

[0027] FIG. **5** illustrates the PCB **112** secured to the base **104** of the housing **102** in accordance with an embodiment. The PCB **112** may also be secured to the connector support **108** of the housing **102**. The PCB may be secured to the connector support **108** and to the base **104** of the housing **102** via the fasteners **118**. The perimeter of the PCB **112** may be adjacent to the first gasket **122** without overlapping the first gasket **122**. In one example, the first gasket **122** completely surrounds an outer edge or outer periphery **123** of the PCB **112**. The base **104** of the housing **102** may include apertures (**128a-128d**) (or “**128**”) that are arranged to receive a fastener (not shown) for fastening or attaching the electronic unit **100** to a panel (not shown), for example, in the vehicle.

[0028] FIG. **6** illustrates a top view of the base **104** of the housing **102** with a second gasket **125** being dispensed onto the base **104**. A top surface **130** of the base **104** includes the divider **132** projecting outwardly from the top surface **130** of the base **104** toward the cover (**106** not shown). As depicted, the divider **132**, along with the perimeter of the base **104**, define the first recess **114** and the second recess **116**. The outer perimeter **134** of the second recess **116** includes projections **136a** and **136b** (“**136**”) extending outwardly from the top surface **130** of the base **104**. The second gasket **125** may be dispensed onto the outer perimeter **134** of the second recess **116** and onto the projections. The base **104** of the housing **102** includes projections **142a-e** (“**142**”) extending outwardly therefrom, and positioned adjacent to the outer perimeter **134** of the second recess **116**. Each projection **142** may be laterally positioned and spaced apart from one another. Each projection **142** may be aligned with a corresponding projection **147a-147e** (“**147**”) (e.g., see FIG. **8**) on the connector support **108**. In one example, the projections **142** may be made from metal.

[0029] FIG. **7** illustrates a top view of the electronics unit **100** including the connector support **108** with a first rib **146** for promoting the sealing of the connectors **110**. The first rib **146** may extend outwardly from a bottom surface **148** of the connector support **108** of the housing **102**. The first rib **146** may be press fit into the second gasket **125** to increase the effectiveness of the seal. In an example, the first rib **146** may be made of metal. The engagement of the first rib **146** into the second gasket **125** may prevent the passage of fluid into the housing **102**.

[0030] FIG. **8** illustrates a more detailed view of the connector support **108** of the housing **102** and the connectors **110**. Each connector **110** includes a head portion **150** and a body portion **152**. The first rib **146** may extend outwardly from the bottom surface **148** of the connector support **108** along a first axis **111**. The first rib **146** may extend laterally along the bottom surface **148** of the connector support **108**. The connector support **108** may include projections **147a-e** (“**147**”) extending outwardly from the bottom surface **148** of the connector support **108** along a second axis **115**. As noted above, in one example, there is a projection **147** aligned with each connector **110** on the connector support **108**. Each projection may include a second rib **154a-e** (“**154**”). Each of the second ribs **154** may extend outwardly from a bottom surface of each projection **147** along the first axis **111**. Each of the second ribs **154** may align with the corresponding projections **142** on the base **104**. In an example, the second ribs **154** may be made of metal. Each of the second ribs **154** may bite into or be press fit (or compressively engage) into the corresponding projections **142** on the

base **104** of the housing **102**. The engagement between the second ribs **154** and the corresponding projections **142** on the base **104** may provide metal to metal contact to promote EMI contact (or ground contact for EMI generated by the electronic unit **100**) for the electronics **113** positioned on the PCB **112**.

[0031] FIG. **9** illustrates the manner which the housing **102** is assembled. For example, the housing includes the base **104**, the cover **106**, and the connector assembly **156** in accordance with an embodiment. First, the housing **102** may be formed as three separate parts (the base **104**, the cover **106** and the connector assembly **156**). In an example, the cover **106** for the housing **102** may be formed using a die cast aluminum or steel. The first gasket **122** may be dispensed onto the bottom surface **124** of the cover **106**. In an example, the first gasket **122** may be dispensed onto the bottom surface **124** of the cover **106** using, for example, a CNC machine. The first gasket **122** may be fitted along an interior of the perimeter of the cover **106**. In addition to being dispensed along the perimeter of the bottom surface **124** of the cover **106**, the first gasket **122** may be dispensed along the perimeter of apertures **120** (not shown) through which fasteners **118** (not shown) may be inserted. The connector assembly **156** may be formed by forming the connector support **108** of the housing **102**. The connector support **108** of the housing **102** may be formed using die casting. In an example, the connector support **108** may be formed from die cast aluminum or steel. The connector support **108** may include the connectors **110**. The PCB **112** may be secured to the connector support **108** of the housing. The connector support **108** may include the first rib **146**. The first rib **146** may extend outwardly from a bottom surface **148** of the connector support **108**. The first rib **146** may align with the second gasket **125** on the base **104** of the housing **102**. When the connector assembly **156** is assembled onto the base **104**, the first rib **146** may press into the second gasket **125** to promote sealing against fluid entering the housing **102**. The second rib **154** on each of the projections **147** on the connector support **108** may align with the projections **142** on the base **104**. The second rib **154** may press into the projection **142** when the connector assembly **156** is assembled onto the base **104**. The first rib **146** may have a height and width that is greater than that of the second rib **154**. The first rib **146** and the second ribs **154** may be axially spaced apart from one another and extend along a generally linear axis over the base **106**. The base **104** may be formed by using die casting. The second gasket **125** may be dispensed onto the top surface **130** of the base **104**.

[0032] FIG. **10** illustrates another view in which the base **104**, the cover **106**, and the connector assembly **156** are assembled with one another to form the housing **102**. As depicted in FIG. **10**, the connector assembly **156** may be aligned with the base **104** so that the first rib **146** may be inserted into (or received by) the second gasket **125**. Each of the second ribs **154** aligns with the corresponding projection **142** on the base **104** so that each of the second ribs **154** may bite or press into the corresponding projection **142**. As depicted in FIG. **10**, the connector assembly **156** is aligned with the base **104** but has not yet been pressed into its final position.

[0033] FIG. **11** illustrates another view in which the **104**, **106**, and **156** are assembled with one another to form the housing **102**. After the connector assembly **156** is aligned with the base **104**, fasteners **118** (not shown) may be delivered to the apertures in the connector assembly **156** and the base **104** and tightened to secure the connector assembly **156** to the base **104**. Then, the cover **106** may be aligned with the connector assembly **156** and the base **104**. Fasteners **118** (not shown) may then be delivered to the apertures **120** in the cover **106** and tightened to secure the cover **106** to the connector assembly **156** and the base **104**. As the fasteners are tightened, the cover **106** and the base **104** are pressed together, sealing the perimeter around the connector assembly **156** and the perimeter around the housing **102**. As depicted in the zoomed-in view of the fit between the connector assembly **156** and the base **104** of the housing **102**, when the connector assembly **156** is assembled onto the base **104** of the housing **102**, the first rib **146** may press into the second gasket **125** so that the second gasket **125** is fitted flush along the bottom surface **148** of the connector support **108**. The flush fit provides a fluid seal that prevents fluid from entering the housing around

the area of the connectors **110**. The second rib **154** may press partially into the corresponding projection **142** so that a recess is left on either side of the second rib **154** between the second rib **154** and the bottom portion of the projection **147**. The first rib **146** is pressed into the second gasket **125** to promote sealing against fluid entering the housing. The second rib **154** bites or presses partially into the corresponding projection **142** to create metal to metal contact for promoting EMI contact between the connector assembly **156** and the base of the housing **104**. The seal provided by the first gasket **122** may provide robust EMI shielding and also water resistance that may achieve an IPX8 rating. The seal provided by the second gasket **125** and the first **146** rib on the connector support **108** may provide water resistance that may achieve an IPX8 rating. The second rib **154** may create metal to metal contact to promote EMI contact. Together, the first gasket **122**, second gasket **125**, first rib **146**, and second rib **154** of the housing **102** provide an electronics unit **100** that promotes water resistance, EMI shielding, and EMI contact.

[0034] While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms of the invention. Rather, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the invention. Additionally, the features of various implementing embodiments may be combined to form further embodiments of the invention.

Claims

1. An electronic unit comprising: a printed circuit board (PCB); a plurality of electronics positioned on the PCB; a housing including a cover and a base that define an opening for receiving the PCB and the plurality of electronics; and a first gasket encircling an outer periphery of the PCB and being configured to at least partially remove electromagnetic interference generated from the plurality of electronics and to prevent liquid from exterior of the housing from penetrating into the opening of the housing.
2. The apparatus of claim 1, wherein the housing includes a connector support to receive a plurality of connectors, the connector support being positioned between the cover and the base and further being positioned on a side of the housing.
3. The apparatus of claim 2, wherein the base of the housing includes a second gasket positioned on an outer perimeter of the base on the side of the housing.
4. The apparatus of claim 3, wherein the connector support includes a first rib extending outwardly therefrom to compressively engage the second gasket to prevent the liquid from exterior of the housing from penetrating into the opening of the housing.
5. The apparatus of claim 4, wherein the connector support further includes a plurality of first projections extending outwardly therefrom, and wherein each of the plurality of projections includes a second rib extending outwardly therefrom to engage the base of the housing to at least partially remove the electromagnetic interference generated by the plurality of electronics.
6. The apparatus of claim 5, wherein the base of the housing includes a plurality of second projections extending outwardly therefrom to engage each of the second ribs provide a conductive path to at least partially remove the electromagnetic interference.
7. The apparatus of claim 1, wherein the first gasket is made from a composition of silicone and metal to at least partially remove the electromagnetic interference.
8. The apparatus of claim 1, wherein the first gasket is dispensed in liquid form onto the cover of the housing and cured in place.
9. An electronic unit comprising: a printed circuit board (PCB); a plurality of electronics positioned on the PCB; a housing including a cover and a base that define an opening for receiving the PCB and the plurality of electronics; and a first gasket positioned on an outer periphery of the PCB and being configured to at least partially remove electromagnetic interference generated from the

plurality of electronics and to prevent liquid from exterior of the housing from penetrating into the opening of the housing.

10. The apparatus of claim 9, wherein the housing includes a connector support to receive a plurality of connectors, the connector support being positioned between the cover and the base and further being positioned on a side of the housing.

11. The apparatus of claim 10, wherein the base of the housing includes a second gasket positioned on an outer perimeter of the base on the side of the housing.

12. The apparatus of claim 11, wherein the connector support includes a first rib extending outwardly therefrom to compressively engage the second gasket to prevent the liquid from exterior of the housing from penetrating into the opening of the housing.

13. The apparatus of claim 12, wherein the connector support further includes a plurality of first projections extending outwardly therefrom, and wherein each of the plurality of projections includes a second rib extending outwardly therefrom to engage the base of the housing to at least partially remove the electromagnetic interference generated by the plurality of electronics.

14. The apparatus of claim 13, wherein the base of the housing includes a plurality of second projections extending outwardly therefrom to engage each of the second ribs to provide a conductive path to at least partially remove the electromagnetic interference.

15. The apparatus of claim 9, wherein the first gasket is made from a composition of silicone and metal to at least partially remove the electromagnetic interference.

16. The apparatus of claim 9, wherein the first gasket is dispensed in liquid form onto the cover of the housing and cured in place.

17. An electronic unit comprising: a printed circuit board (PCB); a plurality of electronics positioned on the PCB; a housing including a cover and a base that define an opening for receiving the PCB and the plurality of electronics; and a connector support to receive a plurality of connectors, the connector support being positioned between the cover and the base and further being positioned on a side of the housing, wherein the cover of the housing includes a first gasket positioned on an outer periphery of the PCB and being configured to at least partially remove electromagnetic interference generated from the plurality of electronics and to prevent liquid from exterior of the housing from penetrating into the opening of the housing, and the base of the housing includes a second gasket positioned on an outer perimeter of the base on the side of the housing, the second gasket being configured to prevent liquid from exterior of the housing from penetrating into the opening of the housing.

18. The apparatus of claim 17, wherein the connector support includes a first rib extending outwardly therefrom to compressively engage the second gasket to prevent liquid from exterior of the housing from penetrating into the opening of the housing.

19. The apparatus of claim 18, wherein the base of the housing includes a plurality of second projections extending outwardly therefrom to engage each of the second ribs to provide a conductive path to at least partially remove the electromagnetic interference.

20. The apparatus of claim 19, wherein the first rib and the second rib are axially spaced apart from one another and extend along a generally linear axis over the base of the housing, and wherein the height and the width of the first rib are greater than the height and the width of the second rib.
